RACHIT KUMAR SINGH

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Professional Summary

Computer Science undergraduate with hands-on experience building a Java 21 distributed commit log and a Python/FastAPI vector database. Applies software engineering principles through fault-tolerance testing, automated quality gates, performance profiling, and cross-functional delivery; brings programming fundamentals across Java, Python, and C++.

Education

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar, Odisha July 2023 – Expected May 2027
B.Tech in Computer Science and Engineering (Currently Enrolled) | CGPA: 8.31/10 | Zero Active Backlogs
Vision International School | Class XII: 84.4% | Class X: 89.8%
University curriculum permits a full-time, 6-month industrial internship semester beginning January 2027.

Technical & Soft Skills

Programming Languages: Java 21, Python, C++, TypeScript, SQL, Object-Oriented Design, Data Structures & Algorithms
Software Engineering: Software Engineering Principles, Software Development Lifecycle (SDLC), Code Reviews, System Design, Distributed Systems, Fault Tolerance
Backend & Web Systems: FastAPI, RESTful APIs, Java NIO, React, Next.js, Microservices
Testing & DevOps: JUnit 5, pytest, Docker, Git, GitHub Actions, CI/CD, Static Analysis & Security Scans
Collaboration & Process: Cross-Functional Teams, Problem-Solving, Root-Cause Analysis, Software Design Documents (SDDs), JIRA

Internship Experience

Accenture – Associate Software Engineering Intern, Bengaluru May 2026 – July 2026
Project: Specification Driven Development (SDD) in Application Management Services (AMS)

- **Accelerated incident response metrics by 40%** and eliminated manual overhead by engineering an automated L1 triage pipeline using a FastAPI RAG chatbot architecture over an indexed vector knowledge base.
- **Orchestrated continuous integration pipelines** across multi-tenant environments, managing full-lifecycle code reviews and pull requests that maintained zero production deployment regressions.
- **Mitigated workflow bottlenecks** by designing an automated root-cause isolation system with statistical confidence scoring, diagnosing 3 structural systemic dependencies to improve process predictability.
- **Optimized cross-functional delivery velocity across 4 distinct engineering teams**, establishing standard technical governance structures and automation rules to improve cross-team SLA compliance by 25%.
- **Authoritative designer of cross-track system blueprints; authored 2 foundational Software Design Documents (SDDs)** adopted by engineering leads to establish technical governance for knowledge storage.

Technical Projects

LogForge: Distributed Commit Log GitHub | Live
Java 21, Java NIO, Object-Oriented Design, Maven, JUnit 5, Docker, GitHub Actions

- **Proved zero acknowledged-record loss** and deterministic fault tolerance across 10 SIGKILL leader failovers covering 1.48M records acked at acks=all by implementing leader-follower replication with ISR tracking, high watermarks, and leader-epoch truncation.
- **Eliminated false failovers** from 22 of 22 runs to 0 of 22 by tracing a blocking socket close that stalled heartbeats about 7s and adding automated regression tests.
- **Engineered a Kafka-style broker** in Java 21 on the JDK alone (about 11K lines) applying core software engineering principles, segmented append-only storage, CRC-32C checksums, torn-write recovery, and idempotent producers.
- **Enforced code review discipline** by gating every pull request and merge on 126 automated JUnit 5 tests with code-coverage thresholds wired into GitHub Actions CI.

VectorForge: Vector Database from First Principles GitHub | Live
Python, NumPy, FastAPI, TypeScript, Next.js, React, Docker, GitHub Actions

- **Delivered 976 queries/s** at 0.954 recall@10 on 200k SIFT1M vectors (42.8x faster than exact search) by implementing HNSW indexing with heuristic neighbour selection, tombstone deletes, and filtered traversal.
- **Cut vector storage 4x** while holding 0.9766 recall@10 on SIFT1M by implementing per-dimension int8 scalar quantization over IVF indexes and validating 100/100 crash-recovery kills.
- **Profiled execution bottlenecks** against FAISS (200k vectors, 1,000 queries, single thread), isolating cache-locality and memory-layout constraints behind a 9.5x matched-recall latency gap.
- **Enforced code quality and security** on every code review and push with 206 engine tests (95.36% coverage) and 30 frontend tests, automating ruff, black, mypy, pytest, pip-audit, and full-history gitleaks scans in CI.

Certifications, Achievements & Activities

- **Competitive Programming:** Solved 300+ DSA and SQL problems on LeetCode (60+ medium, 20+ hard).
- **SOF IMO:** Secured 3 Gold Medals (2014–2016) and 2 Silver Medals (2017–2018) across five sequential national olympiad cycles.
- **Basketball Achievement:** Finished in the top 16 among 100+ registered teams in the XIM Open Tournament.
- **IEEE CTSoc & CIS:** Contributed as a Technical Volunteer for the 23rd OCIT International Conference at KIIT.

Certifications: Accenture Summer Internship Badge and Certification | Microsoft Azure Fundamentals (AZ-900)